Continuous Internal Assessment - 1

Course code: MLA0202
Course: Fundamentals of Machine Learning

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1) Given:

$P(C) = 0.03$

$P(\bar{C}) = 0.97$

$P(+|C) = 0.90$

$P(-|C) = 0.94$

$P(+|\bar{C}) = 0.06$

$P(+) = P(+|C) \cdot P(C) + P(+|\bar{C}) P(\bar{C})$

$= (0.90)(0.03) + (0.06)(0.97)$

$= 0.027 + 0.0582$

$P(+) = 0.0852$

$P(-) = 1 - P(+)$

$P(-) = 0.9148$

$$P(C|+) = \frac{P(+|C)P(C)}{P(+)}$$

$$= \frac{0.90 \times 0.03}{0.0852}$$

$P(C|+) = 0.3169$

$$P(C|-) = \frac{P(-|C)P(C)}{P(-)} = \frac{0.10 \times 0.03}{0.9148} = 0.00328$$

a) $R(T|+) = 2P(\bar{C}|+) = 2(1 - 0.3169)$

$R(T|+) = 1.3662$

b) $P(NT/+) = 50\,P(C/+) = 50\,(0.3169) = 15.8451$

Minimum Risk decision:

Treat if the test result is positive

a) $R(T/-) = 2P(C/-) = 2(1 - 0.00328) = 1.9934$

$R(T/-) = 1.9934$

$R(NT/-) = 50\,P(C/-) = 50\,(0.00328) = 0.16397$

Treat if the test result is negative

$R = P(+)\,R(+) + P(-)\,R(-) = (0.0852)\,(1.3662) + (0.9148)\,(0.16397)$

$\Rightarrow R = 0.1164 + 0.1500$

$R = 0.2664$ units

$$P(C/++) = \frac{P(++/C)\,P(C)}{P(++/C)\,P(C) + P(++/\bar{C})\,P(\bar{C})}$$

$$= \frac{(0.9)^2\,(0.03)}{(0.9)^2\,(0.03) + (0.06)^2\,(0.97)}$$

$$= \frac{0.0243}{0.027792}$$

$\therefore \boxed{P(C/++) = 0.8744}$

2)

For X:

$P(\text{Low}) = 0.30 + 0.02 = 0.32$

$P(\text{Medium}) = 0.28 + 0.10 = 0.38$

$P(\text{High}) = 0.10 + 0.20 = 0.30$

For Y:

$P(\text{NO}) = 0.30 + 0.28 + 0.10 = 0.68$

$P(\text{Yes}) = 0.02 + 0.10 + 0.20 = 0.32$

Entropy:

$H(X) = -\sum P(x) \log_2 P(x)$

$= -[0.32 \log_2 0.32 + 0.38 \log_2 0.38 + 0.30 \log_2 0.30]$

$H(X) = 1.5813$ bits

$H(Y) = -[0.68 \log_2 0.68 + 0.32 \log_2 0.32]$

$H(Y) = 0.9044$ bits

Join Entropy:

$H(X,Y) = -\sum P(x,y) \log_2 P(x,y)$

$= -[0.30 \log_2 0.30 + 0.02 \log_2 0.02 + 0.28 \log_2 0.28 + 0.10 \log_2 0.10 + 0.10 \log_2 0.10 + 0.20 \log_2 0.20]$

$H(X,Y) = 2.0869$ bits

Conditional Entropy:

$H(Y/X) = H(X,Y) - H(X)$

$= 2.0869 - 1.5813 = 0.5056$ bits

Mutual Information $I(X;Y)$

$I(X;Y) = H(Y) - H(Y/X)$

$= 0.9044 - 0.5056$

$= 0.3988$ bits

Cross-Entropy $H(P_x, Q)$

$P_x = (0.32, 0.38, 0.30)$

$Q = (0.5, 0.3, 0.2)$

$$H(P_x, Q) = -\sum P(x) \log_2 Q(x)$$

$$= -[0.32 \log_2 0.5 + 0.38 \log_2 0.3 + 0.30 \log_2 0.2]$$

$$H(P_x, Q) = 1.6531 \text{ bits}$$

KL divergence

$$D_{KL}(P_x \| Q) = H(P_x, Q) - H(P_x)$$

$$= 1.6531 - 1.5813 = 0.0718 \text{ bits}$$

$$P(\text{Yes} / \text{High}) = \frac{P(\text{High}, \text{Yes})}{P(\text{High})} = \frac{0.20}{0.30} = 66.67\%$$

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3) (i) $\hat{\mu} = \frac{\sum x_i}{n} = \frac{48+52+55+47+53+51}{6} = 51$

$$\hat{\sigma}^2 = \frac{\sum (x_i - \hat{\mu})^2}{n} = \frac{9+1+16+16+4+0}{6} = 7.667$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{46}{5} = 9.2$$

$Z = \frac{X - \mu}{\sigma}$, $\sigma = \sqrt{7.667} = 2.769$

$$P(X > 55) = P\left(Z > \frac{55-51}{2.769}\right) = P(Z > 1.444) \approx 0.0743$$

$$P(48 < X < 54) = P\left(\frac{48-51}{2.769} < Z < \frac{54-51}{2.769}\right)$$

$$= P(-1.083 < Z < 1.083)$$

$$\approx 0.721$$

(ii)

Model, $y = w_0 + w_1 x$

$n = 5, \quad \sum x = 150, \quad \sum y = 255$

$\sum x^2 = 5500, \quad \sum xy = 8850$

$$\sum y = n w_0 + w_1 \sum x$$

$$\sum xy = w_0 \sum x + w_1 \sum x^2$$

$$255 = 5w_0 + 150 w_1$$

$$8850 = 150 w_0 + 5500 w_1$$

$w_0 = 11.3, \quad w_1 = 1.33$

$y = 11.3 + 1.33x$

x	Actual y	Predicted y
10	25	24.6
20	38	37.9
30	52	51.2
40	61	64.5
50	76	77.8

$$SSE = \sum (y_i - y_i')^2 = 0.16 + 0.01 + 0.64 + 12.25 + 1.44 = 14.5$$

$$R^2 = 1 - \frac{SSE}{SST} \quad SST = 1790$$

$$R^2 = 1 - \frac{14.5}{1790} = 0.9919$$

Prediction at 60% load

$$y = 11.3 + 1.33(60) = 91.1 \text{ ms}$$

Ridge regularisation: ($\lambda = 500$)

$\bar{x} = 30$

$$\Sigma (x - \bar{x})^2 = 1000$$

$$\Sigma (x - \bar{x})(y - \bar{y}) = 1330$$

Ridge slope:

$$w_1 = \frac{1330}{1000 + 500} = 0.8867$$

$$w_0 = \bar{y} = 51$$

$$y = 51 + 0.8867(x - 30)$$

At $x = 60$

$$y = 51 + 0.8867(30) = 77.6 \text{ ms}$$

4) Total emails = 100

Spam = 40

Legitimate = 60

a) $P(S) = \frac{40}{100} = 0.4$

$P(L) = \frac{60}{100} = 0.6$

b) Spam:

$P(\text{offer}|S) = \frac{28+1}{40+2} = \frac{29}{42} = 0.6905$

$P(\text{free}|S) = \frac{24+1}{42} = 0.5952$

$P(\text{meeting}|S) = \frac{4+1}{42} = 0.1190$

Legitimate:

$P(\text{offer}|L) = \frac{6+1}{62} = 0.1129$

$P(\text{free}|L) = \frac{3+1}{62} = 0.0645$

$P(\text{meeting}|L) = \frac{33+1}{62} = 0.548$

Scam score

$S = P(S) \times P(\text{offer}|S) \times P(\text{free}|S) \times P(\text{meeting}|S)$

$= 0.4 \times 0.6905 \times 0.5952 \times 0.8810$

$S = 0.14483$

Legitimate

$$\lambda = 0.6 \times 0.1129 \times 0.0645 \times (1 - 0.5484) = 0.001974$$

$$P(S|X) = 0.9866$$

c) FP cost = 5

FN cost = 1

$$\text{Threshold} = \frac{FP}{FP+FN} = \frac{5}{5+1} = \frac{5}{6} = 0.833$$

∴ It is spam

d)

$$\text{Accuracy} = \frac{TP+TN}{200} = \frac{45+130}{200} = 87.5\%$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{45}{55} = 81.82\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{45}{60} = 75\%$$

$$\text{F1-score} = \frac{2 \times \text{precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 78.26\%$$

Total misclassification cost

$$= 5(FP) + 1(FN)$$

$$= 5(10) + 15$$

$$= 65 \text{ Units}$$

5) Sigmoid function:

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$net_{h_1} = 0.5(0.4) + 1(-0.2) + 0.1 = 0.1$$

$$h_1 = 0.5250$$

$$net\ h_2 = 0.5(0.3) + 1(0.5) - 0.1 = 0.55$$

$$h_2 = 0.6341$$

Output and error:

$$net_y = 0.6(0.525) - 0.4(0.6341) + 0.2$$

$$y = 0.5650$$

$$E = \frac{1}{2}(1-y)^2$$

$$E = 0.09463$$

Output:

$$\delta_y = (y-t)\,y(1-y) = -1.10692$$

$$\delta h_1 = h_1(1-h_1)(0.6)\,\delta y = -0.01600$$

$$\delta h_2 = h_2(1-h_2)(-0.4)\delta y = 0.00992$$

$$w_{new} = w_{old} - \eta\,\delta x$$

Parameter	Updated value
w_{11}	0.4040
w_{21}	−0.1920
b_1	0.1080
w_{12}	0.2975
w_{22}	0.4950
b_2	−0.1050
v_1	0.6281
v_2	−0.3661
b_3	0.2535

Second forward pass

$h'_1 = 0.5295$, $h'_2 = 0.6315$

$y' = 0.5878$

$E' = \frac{1}{2}(1 - 0.5878)^2$

$= 0.08496$

27/6/23

Percentage error reduction

$$\frac{E - E'}{E} \times 100 = \frac{0.09463 - 0.08496}{0.09463} \times 100$$

$= 10.21\%$